$$\rho_{\mathcal{S}}^{(01)}(t) = \langle 0| \operatorname{Tr}_{\mathcal{E}} \left[\hat{V}(t)\hat{\rho}(0)\hat{V}^{-1}(t)\right] |1\rangle ,$$
$$\rho_{\mathcal{S}}^{(10)}(t) = \langle 1| \operatorname{Tr}_{\mathcal{E}} \left[\hat{V}(t)\hat{\rho}(0)\hat{V}^{-1}(t)\right] |0\rangle . \tag{5.116}$$